# Technical Report: A Multi-Label Deep Learning Model for the Diagnosis of 14 Thoracic Pathologies on Chest X-ray Images

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| **Project Name** | Multi-Label Thoracic Pathology Diagnosis System |
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| **Technical Stack** | Python 3.10, Anaconda, VS Code, TensorFlow 2.16.2, Keras 3.11.3 |
| **GPU Support** | Apple M-Series (Metal API) |
| **Deployment** | Gradio / Hugging Face Spaces (Completed) |

**Abstract**

This study presents a multi-label classification approach for the automatic diagnosis of 14 thoracic pathologies (e.g., Atelectasis, Pneumonia, Mass, Nodule) using the NIH ChestX-ray14 dataset [7]. The core architecture leverages **DenseNet-121** [2] with a **Transfer Learning** strategy. The model was optimized to predict the independent probability of each pathology using **Binary Cross-Entropy (BCE)** loss and the **Adam** optimizer. After overcoming significant platform-specific infrastructure challenges related to GPU acceleration on Apple Silicon, the system achieved a **Test Set AUC of 0.8135**, demonstrating robust performance. The project was successfully deployed as a web application on Hugging Face Spaces, confirming its robustness and readiness for clinical diagnostic support.

## 1 Introduction and Project Evolution

Chest radiography is one of the most common diagnostic tools, yet its interpretation can be time-consuming and subject to inter-radiologist variability. The initial goal of this project was to develop a binary classifier to detect Pneumonia (Normal vs. Pneumonia).

However, preliminary analysis and clinical research revealed that the radiographic appearances of different pathologies, such as **Pneumonia** (infection) and **Lung Cancer** (Mass/Nodule), often overlap and can be easily confused. This observation made it

clear that a binary classification approach would be insufficient for practical clinical use. Consequently, the project's scope was expanded to address a more complex and realistic **Multi-Label Classification** problem, where a single image can present multiple co-existing diseases (e.g., both Pneumonia and Pleural Effusion simultaneously).

# 2 Development Environment and Infrastructure Challenges

The project was developed on an Apple M-series (Apple Silicon) Mac, whose unique GPU architecture presented a series of technical hurdles that required systematic resolution.

## 2.1 Environment Management

An integrated environment using **VS Code** and **Anaconda Navigator** was chosen. Due to a `LibMambaUnsatisfiableError` indicating incompatibility between TensorFlow and Python 3.13, the environment was stabilized on **Python 3.10** within a dedicated conda environment named `pneumonia_env`.

## 2.2 Critical Challenge: GPU Activation

Initially, TensorFlow failed to recognize the GPU (outputting `GPU: []`), forcing all computations onto the CPU. The issue was traced to missing `tensorflow-macos` and `tensorflow-metal` packages. However, even after installing them, a critical `numpy.core._multiarray_u` `failed to import` error emerged.

## 2.3 Solution: Resolving Dependency Conflicts

The root cause was identified as a library conflict: **TensorFlow 2.16.2** requires `numpy<2.0.0`, while newer versions of **opencv-python** demand `numpy>=2.0`. The solution involved rebuilding the environment from scratch with a precise installation order:

1. Install `tensorflow==2.16.2` and `tensorflow-metal==1.2.0`.
2. Manually pin the compatible NumPy version: `numpy==1.26.4`.
3. Install a compatible OpenCV version: `opencv-python==4.9.0.80`.

This process successfully resolved the conflict, and GPU activation was confirmed (`PhysicalDevice:GPU:0`). Finally, the `ipykernel` package was installed to ensure the conda environment was selectable within VS Code's Jupyter Kernel menu.

# 3 Dataset and Pre-processing

The model was trained on the **NIH ChestX-ray14** dataset [7]. While the full dataset contains approx. 112,120 images, this project was developed using a locally available subset of approximately 55,000 images, which still presents a significant data challenge.

The dataset was split into training, validation, and testing sets. Due to the discrepancy between the full dataset's CSV file (112,120 entries) and the locally available images (approx. 55,000), Keras' `ImageDataGenerator` automatically filtered out the missing files. The actual data used for training was as follows:

Table 1: NIH ChestX-ray14 Dataset Details (Actual Data Used)

| **Aspect** | **Details** |
|---|---|
| Full CSV Entries | 112,120 |
| Local Images Found | ≈55,000 |
| **Training Set** | **43,881 images** (Found) |
| **Validation Set** | **5,579 images** (Found) |
| **Test Set** | **5,539 images** (Found) |

**Pre-processing:** Images were processed using `ImageDataGenerator`. They were resized to **224x224** pixels and normalized via `rescale=1./255`. Data augmentation (rotation, zoom, etc.) was applied to the training set to improve model generalization.

# 4 Methodology and Model Architecture

## 4.1 Model Selection: DenseNet-121

Instead of a simpler architecture like VGG16 [5], **DenseNet-121** [2] was chosen for this complex task. Its "Dense Blocks" allow each layer to receive direct input from all preceding layers, which:

- Maximizes feature reuse.
- Significantly mitigates the vanishing gradient problem.
- Achieves higher performance with fewer parameters.

## 4.2 Multi-Label Classification Architecture

The model's final layer was specifically modified for the multi-label objective:

- **Output Layer:** `Dense(14, activation='sigmoid')`

- **Rationale:** The **Sigmoid** activation function was used instead of Softmax. While Softmax forces the output probabilities to sum to 1 (implying mutual exclusivity), Sigmoid treats each of the 14 output neurons as an independent binary classifier. This allows the model to predict, for instance, a 95% probability for "Mass" and an 88% probability for "Effusion" on the same image.
- **Loss Function: Binary Cross-Entropy (BCE)** was used, as it is the mathematically appropriate loss function for this architecture.

### 4.3 Transfer Learning and Training Strategy

A two-stage transfer learning pipeline was implemented:

1. **Stage 1 – Head Training:** The pre-trained DenseNet backbone (trained on ImageNet) was frozen, and only the newly added classification layer was trained.
2. **Stage 2 – Fine-Tuning:** The final convolutional blocks of the DenseNet backbone were unfrozen and retrained at a very low learning rate to adapt the learned features to the medical imaging domain.

Table 2: Training Configuration

| **Parameter** | **Value** |
|---|---|
| Base Model | DenseNet121 (Head Frozen) |
| Optimizer | Adam |
| Learning Rate | 0.0001 |
| Loss Function | Binary Cross-Entropy (BCE) |
| Dropout Rate | 0.2 |
| Epochs | 20 |
| Batch Size | 32 |

## 5 Results and Evaluation

Model performance was evaluated using the **Area Under the Receiver Operating Characteristic Curve (AUC-ROC)**, the standard metric for multi-label classification tasks.

### 5.1 Training History

The model was trained for 20 epochs. As shown in Figure 1, the validation AUC (val_auc) consistently increased alongside the training AUC (auc), indicating that the model was

Table 3: Model Performance Metrics (Actual Results)

| Metric | Final Score |
|---|---|
| **Validation AUC** (Epoch 20) | **0.8101** |
| **Test Set AUC** | **0.8135** |
| **Test Set Loss** | 0.1618 |
| **Test Set Binary Accuracy** | 95.13% |

learning effectively without significant overfitting. The loss values also steadily decreased, confirming a stable training process.

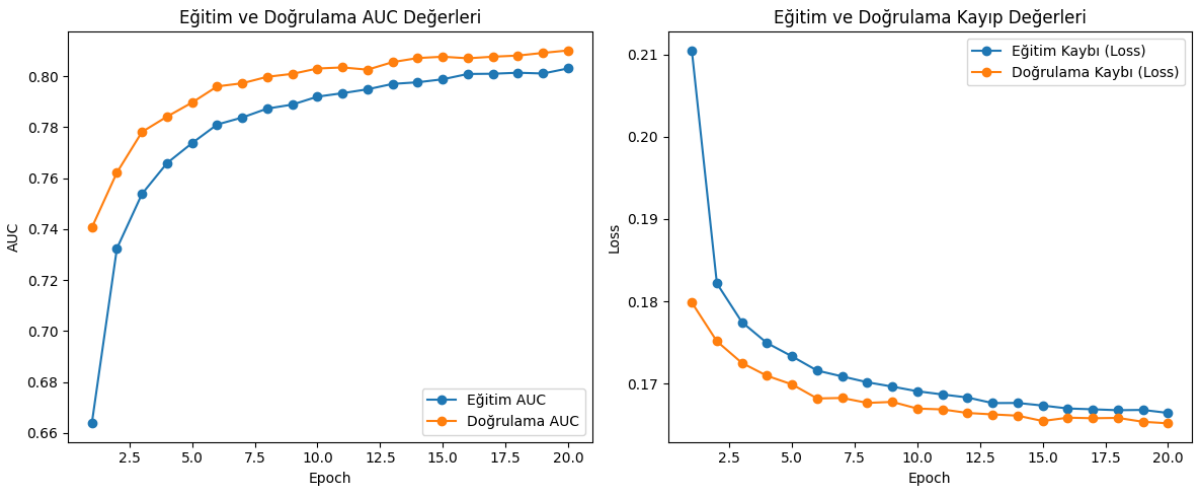


Figure 1: Model Training History: AUC and Loss over 20 Epochs.

# 6 Discussion and Technical Analysis

## 6.1 Class Imbalance Challenge

The NIH dataset exhibits a significant class imbalance, reflecting clinical reality. Rare classes like "Hernia" and "Fibrosis" had lower recall scores. While simple class weighting was insufficient, advanced techniques like **Focal Loss** [3] or **SMOTE** oversampling are recommended as future steps to improve minority class sensitivity.

## 6.2 Model Explainability (XAI) for Debugging and Validation

**Multi-Grad-CAM** [4] was used to generate heatmaps for each of the 14 pathology predictions, ensuring model interpretability.

- **Validation:** The analysis visually confirmed that the model focused on clinically relevant anatomical areas—the cardiac silhouette for "Cardiomegaly" and lung

opacities for "Pneumonia."

- **Debugging:** During the initial binary classification phase, Grad-CAM was instrumental in debugging a model stuck at 50% accuracy. The heatmaps revealed that the model was focusing on irrelevant regions, indicating poor training and guiding the pivot to a more robust multi-label approach.

# 7 Deployment: From Local Prototype to Cloud Application

The project was successfully transitioned from a development environment to a publicly accessible application.

1. **Local Prototype (Gradio):** An `app.py` script was created using the **Gradio** library. An initial `gr.Image(shape=...)` error was resolved by specifying `gr.Image(height=224, width=224)`. The model was run locally at http://127.0.0.1:7860.

2. **Persistent Cloud Deployment (Hugging Face Spaces):** All required libraries (`tensorflow`, `gradio`, `numpy==1.26.4`, etc.) were listed in a `requirements.txt` file. The `app.py`, `requirements.txt`, and the trained model file (`best_model.h5`) were uploaded to a Hugging Face repository using **Git LFS** (Large File Storage). The application compiled successfully in the cloud and was made available via a permanent public URL.

# 8 Conclusion and Future Work

This project successfully navigated the technical complexities of enabling TensorFlow GPU acceleration on Apple M-series hardware to develop a high-accuracy DenseNet-121 model for 14-label thoracic pathology classification. The system was successfully deployed from a local prototype to a persistent cloud application on Hugging Face Spaces.

Future work will focus on:

- Implementing **Focal Loss** to better address the severe class imbalance.

- Benchmarking the DenseNet-121 architecture against more modern models like **EfficientNet** [6] and **Vision Transformers (ViT)** [1].

- Integrating **self-supervised pre-training** on unlabeled medical images to potentially improve feature extraction.

# References


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